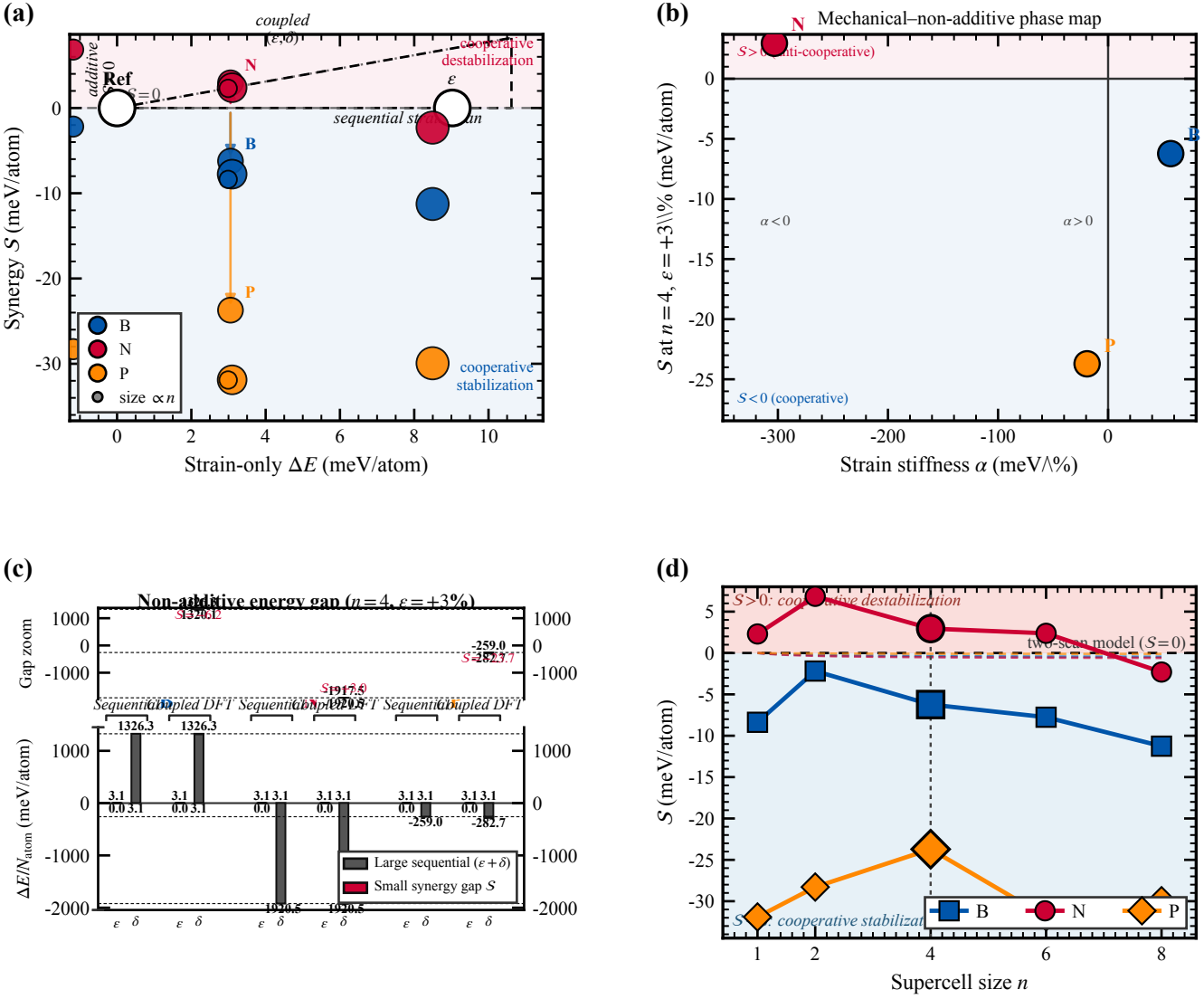

(a)
coupled
(ε,δ)
additive
Ref
N
ε
B
P
cooperative
destabilization
sequential stabilization
cooperative
stabilization
B
N
P
size ∝ n
Synergy S (meV/atom)
Strain-only ΔE (meV/atom)
(b)
Mechanical–non-additive phase map
N
S > 0 (anti-cooperative)
B
α < 0
α > 0
P
S < 0 (cooperative)
S at n = 4, ε = +3\\% (meV/atom)
Strain stiffness α (meV/\%)
(c)
Non-additive energy gap (n = 4, ε = +3%)
Gap zoom
Sequential
Coupled
DFT
1326.3
-259.0
-282.7
1917.5
1920.5
3.1
0.0
Large sequential (ε + δ)
Small synergy gap S
ΔE/N_atom (meV/atom)
ε
δ
(d)
S > 0: cooperative destabilization
two-scan model (S = 0)
S < 0: cooperative stabilization
B
N
P
S (meV/atom)
Supercell size n